Quick Start Guide
Digital Buffer

DB-50-F-A
DB-50-R-A

1 About This Guide

This document is a condensed quick-start guide intended to support basic installation and operation of the device. This guide does **not** replace the full product manual. The complete operation manual can be downloaded at: https://www.flimlabs.com/products/digital-buffer/

Scan the QR code to open the full manual.

Reading the complete manual is strongly recommended before operating the device, unless the user is already familiar with similar laboratory instrumentation.

2 Safety Information

Do not operate the device if it is damaged or if safe operating conditions cannot be ensured.

- Read the full manual before installation and use.
- Operate only within the specified electrical and environmental limits.
- Do not open the enclosure.
- Disconnect power before modifying connections.
- Use only undamaged and properly rated cables.

3 Intended Use

The Digital Buffer is intended as a compact electronic module for the conversion and distribution of digital signals by professional users in private research laboratories, companies, and universities. It may be used as a stand-alone unit or integrated into custom measurement systems and instrumentation.

The device must be operated only by qualified personnel familiar with laboratory electronic instrumentation.

4 Product Overview

The Digital Buffer is a compact electronic module designed for the conversion and distribution of pulse digital signals into standardized digital outputs. It receives a digital input signal via SMA, converts the signal level, and replicates it on two electrically independent SMA output channels while preserving the timing characteristics of the original pulse.

The device is optimized for signal conditioning and distribution in spectroscopy, fluorescence lifetime measurements, and time-correlated photon counting systems.

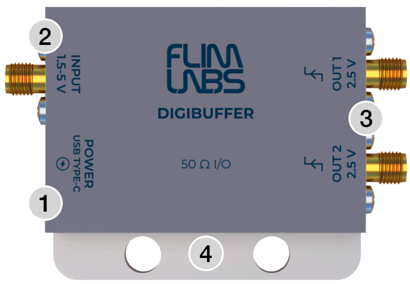
Key Features

- One SMA pulse input
- Two SMA digital outputs
- Standard LVTTL 2.5 V outputs on 50 Ω
- USB Type-C 5 V power supply
- No user configuration required
- Compact laboratory form factor
- Available for falling-edge or rising-edge input signals

5 Components Overview

Overview of the key components of the device.

1. USB Type-C power input
2. SMA signal input
3. SMA output 1 and 2
4. Mounting holes for mechanical fixing



6 Scope of Delivery

- Digital Buffer
- USB Type-C power cable
- Quick start guide
- Declaration of conformity

7 Setup

Install the device on a stable laboratory surface. Before installation:

- inspect the device and accessories for visible damage
- verify connectors and cables are intact
- verify that the input signal meets the requirements given in Table 1

8 Connections

Power connection

- Connect the USB Type-C cable to the device power input.
- Connect the other end to a suitable 5 V DC power source such as a computer USB port, a power bank, or a certified 5 V DC adapter.
- Verify visually that the LED power indicator confirms correct device power-up.

Signal connection

- Connect the digital input signal to the SMA input connector.
- Connect one or both SMA outputs to the downstream measurement or processing devices.

9 Configuration

After completing all electrical connections, no further configuration is required.

Verify that the device powers up correctly and that stable output signals are available on the connected output channels under the expected operating conditions.

10 Operation

Once the power supply, input signal, and output connections have been correctly established, the Digibuffer operates autonomously without further user intervention.

11 Further Information

Detailed information on troubleshooting, maintenance, storage, and disposal is provided in the full operation manual available at https://www.flimlabs.com/products/digital-buffer/.

12 Product Label

Digital Buffer

P/N: DB-50-X-A
S/N: DB-XXXX-XXX
5 V ⎓ 0.2 A | IP40

MADE IN ITALY - **FLIM LABS S.r.l.**
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13 Technical Specifications

Item	DB-50-F-A	DB-50-R-A
Electrical Specifications		
Electrical safety class	Class III	
Power supply	USB Type-C, 5 V DC	
Current consumption (max)	0.2 A	
Power consumption (max)	1 W	
Signal Specifications		
Input connections	1x SMA	
Input polarity	negative	positive
Minimum input pulse width	>4 ns	
Minimum input pulse height (abs)	1.5 V	
Maximum input pulse height (abs)	5 V	
Output connections	2x SMA	
Output signal	SMA: 2.5 V digital LVTTL at 50 Ω	
Output pulse width	>4 ns	
Maximum repetition frequency	100 MHz	
Impedance	50 Ω	
Physical Specifications		
Dimensions (L x W x H)	60 mm x 42 mm x 14 mm	
Weight	80 g	
Protection rating	IP40	
Environmental Specifications		
Operating temperature	0 °C to +40°C	
Storage temperature	-20 °C to +70°C	
Humidity	up to 80% non-condensing	

Table 1: Technical specifications of the Digital Buffer.